

Backbone Chemistry Governs the Reliability of Machine-Learned Glass Transition Temperatures: A Family-Resolved Benchmark with Novelty-Conditioned Conformal Uncertainty

Abstract

Machine-learned models of the polymer glass transition temperature (T_g) routinely report coefficients of determination above 0.85 and are proposed as screening tools, but those figures come from random partitions whose test sets are filled with chemistry already well represented in training. Using 7,174 canonical repeat units in twenty-one backbone families, we measure what deployment outside it costs. For histogram gradient boosting, mean absolute error rises from 28.1 K on random partitions to 33.8 K under scaffold holdout, 44.5 K under fingerprint-cluster holdout and 48.1 K under whole-family holdout, and pooled R^2 falls from 0.87 to 0.75, while the two best models differ by 0.38 K. A matched-pair design, fixing the test set and varying only which structures the training pool loses, attributes +14.97 K (95% CI [+11.21, +19.28]) to removing the family and +0.29 K to removing an equal number of unrelated structures, the family arm being worse in all twenty-one families for both models. Uncertainty fails in the same place: split conformal holds 0.904 marginal coverage on random splits while covering 0.650 of the least familiar structural band, and loses marginal validity under shift. Family-conditioned calibration cannot repair this, since a withheld family has no calibration members; similarity-conditioned calibration holds 0.877–0.907 across all four regimes.

1. Introduction

The glass transition temperature is the single most consulted thermal descriptor of an amorphous polymer. It bounds the service window of a thermoplastic, sets processing conditions, and governs the mechanical response of a matrix in a composite. Measuring it by differential scanning calorimetry or dynamic mechanical analysis is routine but slow, and it requires a synthesised, purified sample, so predicting T_g from the structure of the repeat unit has been a target of polymer theory since group-contribution methods were first formalised [VanKrevelen2009, Bicerano, Askadskii2003].

Data-driven surrogates have largely displaced those schemes. Curated repositories such as PoLyInfo [Otsuka] made supervised learning on experimental T_g practical, and a decade of work has explored the resulting design space: physically interpretable descriptor sets with tree ensembles and kernel methods [Tao2021], graph neural networks over the repeat-unit graph [Qiu2023], and chemical language models pretrained on millions of hypothetical polymers [Kuenneth2023, Xu2023]. Recent community infrastructure has consolidated the datasets and the evaluation code [Jablonka2025, Xu2025]. The accuracies reported are genuinely useful: mean absolute errors in the region of 25–30 K against experimental values whose own inter-source spread can exceed 100 K.

Those numbers describe interpolation. They are obtained on random partitions, in which a test repeat unit typically has close structural analogues in training, and they therefore estimate how well a model fills in gaps within chemistry it has already seen. The motivating application is the reverse. A screening campaign enumerates candidates precisely because they are not yet in the literature, and the candidates that matter most are the ones furthest from the training distribution. The mismatch between the validation regime and the deployment regime is well known in cheminformatics, where scaffold and time-based splits were introduced for exactly this reason [Sheridan2013, Wu2018], and it has begun to be examined for polymers [Babbar2023, Akdogan2026].

Two questions remain open, and this work addresses both.

Is the family-holdout penalty caused by the missing chemistry? Withholding an entire polymer family and observing an error increase is the standard diagnostic, but it changes two things at once. The test population changes — families differ in how intrinsically hard they are and in how wide their T_g distribution is — and the training set shrinks. A comparison against a random-split baseline therefore measures the sum of a population effect, a data-volume effect and the effect of interest. Published treatments of this dataset state the confound explicitly and leave it unresolved. Resolving it requires holding the test set fixed and matching the training pools on size.

Does the uncertainty survive the shift? A point prediction without a trustworthy interval is not actionable for screening, and split conformal prediction [Papadopoulos2002, Vovk2005, Lei2018] is an attractive answer because it wraps any regressor in intervals with a finite-sample coverage guarantee under exchangeability. Two difficulties arise here. Exchangeability is precisely what a family holdout violates. And the guarantee is *marginal*: a predictor can achieve 90% coverage overall while systematically under-covering a subpopulation, and in this problem that subpopulation is the unfamiliar chemistry the model was deployed to explore. A recent analysis on a 410-sample simulation-derived set reports this pattern and characterises conformal intervals as conservative triage indicators rather than fine-grained screening tools [Akdogan2026]; whether the failure is repairable, and whether it persists at the scale and heterogeneity of the experimental record, has not been tested.

1.1. Contributions

- **An independently re-derived, chemically resolved dataset.** We re-curate 7,208 experimental repeat-unit records to 7,174 canonical structures, reproducing the published audit of this collection step for step, and assign each structure to one of twenty backbone families or to an explicit abstention class using a *backbone-aware* substructure classifier. Classifying on the chain path rather than on whole-molecule matches keeps poly(alkyl acrylate)s out of the polyester class, which matters because every family-holdout conclusion inherits the labels.
- **A matched-pair design that isolates the family effect.** Three training pools — one retaining the family, one with it removed, and one size-matched control from which an equal number of unrelated structures was removed — predict an identical held-out test set, separating the cost of missing chemistry from the cost of missing data.
- **A conditional-coverage evaluation of conformal uncertainty,** and a novelty-conditioned Mondrian taxonomy that restores subgroup validity where a family-conditioned taxonomy structurally cannot.
- **A validated applicability-domain criterion** expressed in units a practitioner can act on: the nearest-neighbour Tanimoto similarity below which the stated interval should not be believed.
- **A group-contribution reference point and a taxonomy sensitivity check,** so that the machine-learned result is placed against the additive schemes it proposes to replace and against a perturbation of its own labels.
- **A reproducible open implementation,** with leak-free in-fold preprocessing and a test suite pinning both the chemistry of the family assignment and the finite-sample behaviour of the conformal predictors.

2. Materials and Methods

2.1. Dataset and curation

We use the experimental homopolymer T_g collection redistributed with the POINT² polymer informatics benchmark [Xu2025], which derives from the PoLyInfo record [Otsuka]. Each entry pairs a repeat unit written as a PSMILES string — SMILES with two `\*` atoms marking the polymerisation attachment points — with a reported T_g .

The raw export is record-level: one repeat unit may appear several times carrying values from different primary sources. Modelling on records would allow identical inputs to fall on both sides of a partition and would weight frequently reported polymers more heavily, so all analysis is performed on a structure-level table. Repeat units were canonicalised with RDKit and grouped; where a canonical structure carried several reported values, the median was taken, and the number of contributing records and the full reported range were retained as provenance.

2.2. Backbone-aware family assignment

Polymer family labels were derived rather than inherited, so that the taxonomy is reproducible and its chemistry is inspectable. For each repeat unit the *backbone* is taken as the union of shortest paths between attachment points, extended to include any ring the path enters. Characteristic linkages are then matched against SMARTS patterns and accepted only when their core atoms lie on that backbone.

Because a repeat unit may be written cut at its own characteristic linkage — nylon-6 as *NCCCCC(=O)* divides its amide between the two ends — the classifier operates on a head-to-tail dimer rather than on the single unit. Joining two copies restores the linkage the cut destroyed while keeping the junction a bond that exists in the real chain.

Backbone restriction is what separates a poly(alkyl acrylate) from a polyester: both contain an ester group, but in the acrylate it decorates an all-carbon chain. Patterns are tested in order of specificity, because the groups are nested — every cyclic imide contains two amide-like motifs, and a urethane contains both ester-like and amide-like fragments. Repeat units whose backbone carries no heteroatom linkage are classified by the groups pendant to the chain (aromatic-dominated, acrylic, styrenic, halogenated, other heteroatom-substituted vinyl, unsaturated, or saturated hydrocarbon). A unit whose backbone does carry a heteroatom but matches no linkage pattern is assigned to an explicit abstention class, **Other backbone**, rather than forced into the nearest hydrocarbon category. The twenty-five chemistry cases in Table S1 are pinned by unit tests.

2.3. Representations

Two representations were computed. The full RDKit descriptor block (217 columns) provides physically interpretable features; the topological `Ipc` index was replaced by its base-10 logarithm because it overflows for larger repeat units. Binary Morgan fingerprints (radius 2, 2,048 bits) provide the similarity space in which structural novelty is measured. Attachment points were retained in both, since they identify where the chain continues and the environments around them encode exactly the linkage motifs that set T_g .

Descriptor column filtering — removal of non-finite, constant and exactly duplicated columns — is implemented as the first step of a scikit-learn pipeline and is therefore **fitted on the training fold only**. Performing this filtering once over the full table, as is common, lets test structures influence the feature definition; the closest prior analysis of this dataset notes this as an unquantified limitation of its own results.

2.4. Partitioning regimes

Four regimes were used, in increasing order of enforced novelty: **random** partitions (repeated, 80:20); **scaffold** partitions, in which generic Bemis–Murcko scaffolds [Bemis1996] are assigned whole; **cluster** partitions, using average-linkage agglomerative clusters in Jaccard space over Morgan bits; and **family** holdouts, in which every member of one backbone family is withheld. All non-family regimes were repeated ten times with different seeds, because single-split

rankings among tree ensembles on this dataset fall inside run-to-run noise. The family regime contributes twenty-one splits, one per family.

2.5. Matched-pair design

For each eligible family F , a test set $T \subset F$ is drawn (30% of F) and three training pools are constructed: **informed** (everything but T , so the remainder of F is available), **naive** (everything but F entirely), and **control** (**informed** minus a randomly chosen block of structures from other families, equal in number to $F \setminus T$). All three predict the same T .

The contrast **naive** – **control** is the quantity of interest. Both pools are identical in size and both exclude T ; they differ only in *which* structures were removed. The contrast **control** – **informed** measures the pure data-volume effect over the same test set.

Two properties of the design bound what it can establish. First, the removed block is small: it is 70% of one family, a median of 1.85% of the informed pool across the twenty-one families, ranging from 0.43% for polyureas to 17.94% for polyimides, with sixteen of twenty-one families under 5%. Second, the two arms remove the same *quantity* but not the same *kind* of structure: the naive arm removes a coherent chemical neighbourhood, the control arm a diffuse random sample spread over the remaining families. The contrast therefore measures the effect of removal *locality* at fixed volume. That chemistry is what locality means here is an interpretation, argued in Section 4.2, not a separate measurement.

2.6. Models

Five regressors were evaluated on the descriptor representation: a training-median baseline, support vector regression, random forest, histogram gradient boosting and extremely randomised trees. Every model is wrapped in the same pipeline so that all data-dependent preprocessing is fitted in-fold.

A sixth model, a group-contribution baseline, was run through the same four regimes. It implements the classical Van Krevelen additive form $T_g = Y_g/M$ with $Y_g = \sum n_i Y_i$, where M is the repeat-unit molar mass, n_i the count of structural group i and Y_i its molar glass-transition contribution [VanKrevelen2009, Bicerano, Askadskii2003]. Groups are RDKit fragment counts supplemented by element, ring, rotatable-bond, hydrogen-bonding and backbone-length counts, which carry the chain skeleton that fragment counts alone would miss. The contributions Y_i are fitted by ridge regression on the training fold rather than taken from a published table, so the additive scheme receives the same data the ensembles receive and any difference between them reflects the functional form rather than the calibration set. Fitting is performed in $Y_g = T_g \cdot M$ space and in absolute temperature, since an additive law expressed in Celsius would make the contributions depend on an arbitrary zero.

2.7. Conformal prediction and coverage diagnostics

Each training pool is divided three ways: a **fit** partition (60%) trains the point regressor, a **difficulty** partition (20%) supplies out-of-sample residuals, and a **calibration** partition (20%) supplies the conformal quantile. Fitting the difficulty model on the point model's own training residuals would teach it that the model is everywhere accurate, so that partition must be held out. All conformal variants share one point predictor and one calibration set, so differences between them reflect only how residuals are converted into intervals.

The three-way division costs accuracy. On a random 80:20 partition the point regressor is fitted on about 48% of the dataset, three-fifths of what an 80%-trained literature model sees, so the random-split errors reported below are mildly pessimistic relative to published figures obtained without a calibration carve-out.

Given calibration scores $s_1 \dots s_n$ and a miscoverage level α , the conformal quantile is the $[(n+1)(1-\alpha)]$ -th order statistic; when that index exceeds n the honest interval is unbounded. We report $\alpha = 0.1$ throughout. Three predictors are compared:

- **SCP** — one global quantile of absolute residuals.
- **Normalised SCP** — residuals scaled by a fitted difficulty estimate, so intervals widen where the model expects to struggle rather than everywhere.
- **Mondrian SCP** — a separate quantile per category [Vovk2012], under two taxonomies: *polymer family*, and *nearest-training-similarity band*.

The asymmetry between the two taxonomies is the point. A family-conditioned taxonomy is undefined for a family with no calibration members, which is exactly the family-holdout case; a novelty-conditioned taxonomy is defined for any repeat unit, seen family or not.

A fourth, **width-matched control**, is used only to test whether the novelty-conditioned predictor's advantage reduces to its being wider. It multiplies every split conformal interval about its centre by a single constant k , chosen so that the mean width equals the novelty-conditioned mean width on the same test set. It is an oracle and not a deployable method, because k is read off the test-set widths it is then compared against; its role is to bound what *any* global inflation of split conformal could achieve.

Alongside marginal coverage and mean interval width we report **conditional** coverage within each similarity band and each family, the **maximum coverage gap** (the largest shortfall below nominal across subgroups with at least ten members), and the coverage-width criterion. Ensemble spread is additionally assessed by sparsification error.

2.8. Statistical treatment

Confidence intervals on the matched-pair effects are percentile bootstrap intervals over the twenty-one family-level means. Those twenty-one values are neither independent nor a sample. The arms of a given family share at least 82% of their training data by construction, and more than 95% in sixteen of the twenty-one families; the families are chemically nested, so that polycarbonates sit inside the same carbonyl chemistry as polyesters; and the twenty-one are the entire population of families in this dataset rather than a draw from a larger one. The intervals should therefore be read as descriptive dispersion across families, not as inference about a population of families.

The Wilcoxon signed-rank test on twenty-one paired values has a smallest attainable two-sided p of $2^{-20} = 9.5 \times 10^{-7}$, reached whenever all twenty-one signs agree. Where that value appears below it is the floor of the test, not a measured quantity, and the informative statement is the sign count.

p -values are reported unadjusted for multiplicity. The one place where this matters is the driver analysis of Section 3.3, which runs sixteen correlations (four candidate drivers $\times$ four models). Under a Bonferroni correction at $\alpha = 0.05/16 = 0.0031$, the nearest-training-similarity correlation survives for all four models and no other driver survives for any.

2.9. Reproducibility

All code, the curated structure-level table and every result file are available at <https://github.com/Rugvedhs/spectrasense>, in the directory ``polymer-tg-benchmark``. Seeds are fixed; a cold run reproduces every number in this paper.

3. Results

3.1. The public record survives re-derivation

Independent re-curation of the 7,208 raw records reproduces the published audit of this collection step for step: 7,174 unique raw SMILES, no RDKit parse failures, 7,174 canonical structures, 31 canonical groups containing more than one record, 34 records absorbed by aggregation, 28 of those groups carrying disagreeing T_g values, and a maximum within-structure range of 115 K (Table 1). Attachment-point counts likewise agree, at 7,170 structures with two,

three with three and one with four. The dataset is therefore a stable object to build on, and the numbers below are not sensitive to curation choices.

That 115 K figure deserves emphasis before any model error is quoted. It is the spread between independent literature reports of the *same* repeat unit, and it places a floor under what structure-based prediction can achieve: a repeat-unit representation cannot resolve differences in molecular weight, tacticity, crosslinking, thermal history or measurement protocol, and those differences are folded into the target.

The derived taxonomy places all twenty-five reference polymers of Table S1 in the family a polymer chemist would assign, and yields the family composition of Table 2. The resulting family T_g ordering (Figure 1) is likewise the expected one, which is a further check that the labels are chemically meaningful rather than merely self-consistent: among the named families, polyphosphazenes (median -8°C) and polysiloxanes (-8°C) sit lowest, reflecting the low rotational barriers of P=N and Si-O backbones, and aromatic polyimides sit highest (249°C), reflecting rigid, strongly interacting heteroaromatic chains. The only group below them is the abstention class Other backbone (median -19.6°C , 58 structures), of which 57 contain silicon; its position is consistent with the silane and carbosilane chemistry it holds.

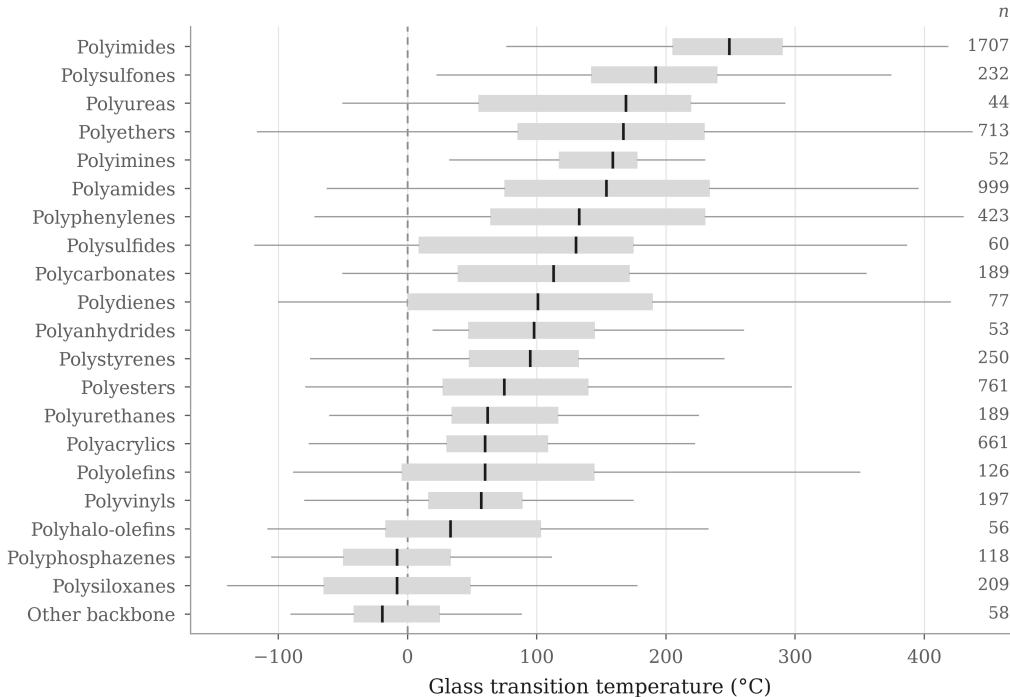


Figure 1. Reported glass transition temperature by repeat-unit family, ordered by median; boxes span the interquartile range, whiskers the 5th to 95th percentiles, and family sizes are listed at the right. The ordering is the one polymer chemistry predicts, which is evidence that the derived taxonomy is meaningful rather than merely self-consistent.

3.2. Accuracy is governed by the split, not by the model

On random partitions the best model, histogram gradient boosting, reaches 28.1 ± 1.0 K MAE ($R^2 = 0.87$), consistent with published values for descriptor-based ensembles on this dataset once the calibration carve-out of Section 2.7 is allowed for. That ranking is fragile. Across ten repeated splits, histogram gradient boosting beats extremely randomised trees by 0.38 K, which a paired t -test does not resolve ($p = 0.069$); random forests and support vector regression are worse by 2.17 K and 2.74 K ($p < 0.0001$ each). Reporting a single split, as is common, would allow either of the leading models to be declared the winner.

Against that sensitivity, the choice of partition moves the error by far more (Figure 2, Table 3). Holding the model at histogram gradient boosting, MAE rises from 28.1 K on random splits to 33.8 K on scaffold splits, 44.5 K on fingerprint-cluster splits and 48.1 K under whole-family

holdout. The random-to-family contrast is therefore 20.0 K: about fifty times the 0.38 K that separates the two best models, and about seven times the 2.74 K that spans all four non-trivial regressors. Only the training-median baseline, at +65.7 K, is moved more by the choice of model than by the choice of protocol. Whatever a leaderboard on this dataset is ranking, it is not the property that changes most.

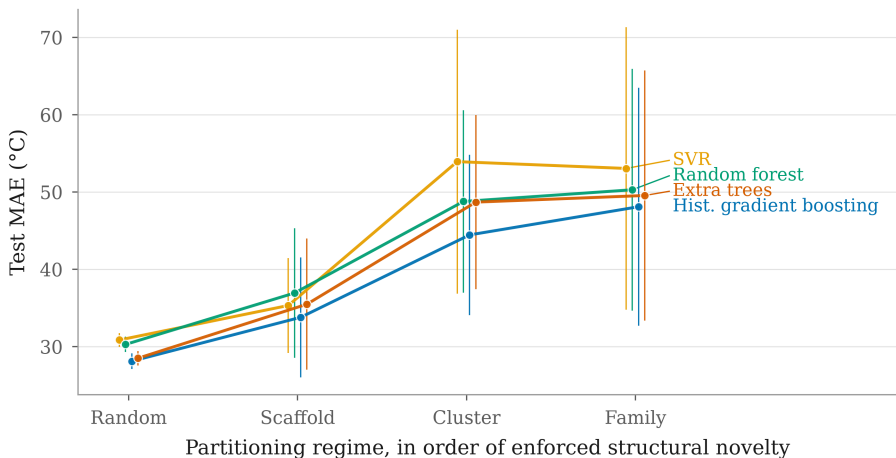


Figure 2. Test MAE by model and split regime; error bars are the standard deviation across repeated splits. The spread between regimes dwarfs the spread between models.

Mean nearest-training Tanimoto similarity falls alongside the error, from 0.76 on random splits to 0.67, 0.40 and 0.45. That ordering is not strictly monotone: family holdout leaves test structures slightly *less* isolated on average (0.45) than cluster holdout (0.40), because a withheld family may still have near analogues in a chemically adjacent one, yet family holdout produces the higher error. Structural distance alone therefore does not determine difficulty; *which* chemistry is missing also matters, a point Section 3.3 takes up directly.

R^2 must be quoted with its aggregation stated, because the two natural choices disagree by more than any effect in this paper (Table 10). Pooling residuals over all held-out structures in a regime, which is the quantity comparable to a random-split R^2 , gives 0.87 on random splits, 0.81 on scaffold, 0.68 on cluster and 0.75 under family holdout. Averaging R^2 over the twenty-one individual family holdouts instead gives 0.26, because each family is scored against its own T_g variance and a family with a narrow T_g spread contributes a large negative value at a modest absolute error. That 0.26 is a real number but it is not commensurable with 0.87, and the fall from 0.87 to 0.26 that a naive comparison suggests is an aggregation artefact. The comparable fall is 0.87 to 0.75.

The dispersion is as informative as the mean. Random-split MAE varies by ± 1.0 K across ten repeats, and scaffold and cluster splits by ± 7.8 K and ± 10.4 K; these are repeat-to-repeat quantities. The ± 15.4 K quoted for family holdout is not: each family is held out exactly once, so it is the standard deviation *between families*, and it describes how unevenly the penalty is distributed over chemistry rather than how reproducible any one holdout is. Either way, a point estimate from one chemistry-aware split should not be trusted.

3.3. Transfer failure is a property of backbone chemistry

Family-holdout deterioration, defined as the ratio of family-holdout MAE to random-split MAE for the same model, spans a factor of nearly three across the twenty-one families (Table 4; histogram gradient boosting throughout this section). At the difficult end sit polyphosphazenes (3.02 $\times$, 84.8 K MAE), polydienes (2.83 $\times$, 79.6 K) and polysiloxanes (2.48 $\times$, 69.7 K). At the easy end, polyvinyls deteriorate by 1.07 $\times$, polyethers by 1.19 $\times$ and polycarbonates by 1.20 $\times$; no family is predicted better when withheld than the average random-split structure, though polyvinyls come within 8%.

The signed error makes the mechanism visible rather than merely plausible. Defining bias as $\text{mean}(T_g^{\text{observed}} - T_g^{\text{predicted}})$, so that a negative value is overprediction, the three families a model has least chemical warrant for are overpredicted by a wide margin when withheld: polyphosphazenes by -83.7 K, polysiloxanes by -63.7 K and the Other backbone silane class by -45.3 K. Set against their MAEs of 84.8, 69.7 and 52.5 K, the error in these families is dominated by displacement rather than by scatter. This is the expected failure for an inorganic backbone withdrawn from a training set dominated by organic condensation polymers: with nothing to indicate that P=N or Si-O rotation is nearly free, the model assigns these chains the stiffness of the organic chemistry it does know. The mirror case exists and points the other way. Polydienes are *under*predicted by $+48.0$ K and polyolefins by $+37.8$ K; a model deprived of its flexible unsaturated and saturated hydrocarbon chains places them too low, having learned its temperature scale from stiffer condensation backbones. Bias of either sign is what an absent linkage type produces, and it is not confined to the extremes: across the twenty-one families the median ratio of $|\text{bias}|$ to MAE is 0.54, so about half of a typical family-holdout error is systematic displacement. Its magnitude tracks structural isolation, the absolute bias correlating with mean nearest-training similarity at $r = -0.53$ ($p = 0.014$, unadjusted).

Polycarbonates illustrate the benign case from the same angle: they are surrounded in structure space by polyesters and polyethers, which remain in training and carry nearly the same carbonyl and ether linkages, so their withdrawal removes little the model cannot recover, and their bias is a modest $+15.8$ K.

Four candidate explanations were tested against deterioration across the twenty-one families, and only structural isolation survives. Mean nearest-training similarity correlates strongly and negatively for every model (Pearson r between -0.62 and -0.69 , $p \leq 0.0027$, significant for all four models after Bonferroni correction over the sixteen tests). The shift between the family's median T_g and the rest of the dataset gives Pearson r between 0.32 and 0.39 ($p \geq 0.079$), and the ratio of the family's T_g interquartile range to the dataset's gives r between 0.15 and 0.32 ($p \geq 0.16$); neither is resolved here. Family size is likewise not resolved, at r between -0.05 and -0.35 ($p \geq 0.115$; for extremely randomised trees, $r = -0.27$, 95% CI $[-0.63, +0.18]$). That interval is wide enough to contain a moderate effect in either direction, so the honest statement is that a size effect is undetectable at twenty-one families, not that it is absent.

One feature of the design protects the headline contrast. Mean nearest-training similarity and family size are nearly orthogonal across the twenty-one families ($r = 0.10$, $p = 0.67$), so the similarity result is not a relabelled size result and the two candidate drivers are not competing for the same variance.

3.4. Conformal intervals lose validity exactly where they are needed

All conformal results below use extremely randomised trees as the point predictor. Split conformal behaves as advertised on random partitions: 0.904 empirical coverage against a nominal 0.90. Under shift the guarantee degrades with structural novelty — 0.834 on scaffold splits, 0.730 on cluster splits and 0.701 under family holdout (Figure 4, left; Table 8). This is not a defect of the method but of its premise: conformal validity requires calibration and test data to be exchangeable, and a chemistry-aware holdout is constructed precisely to break that.

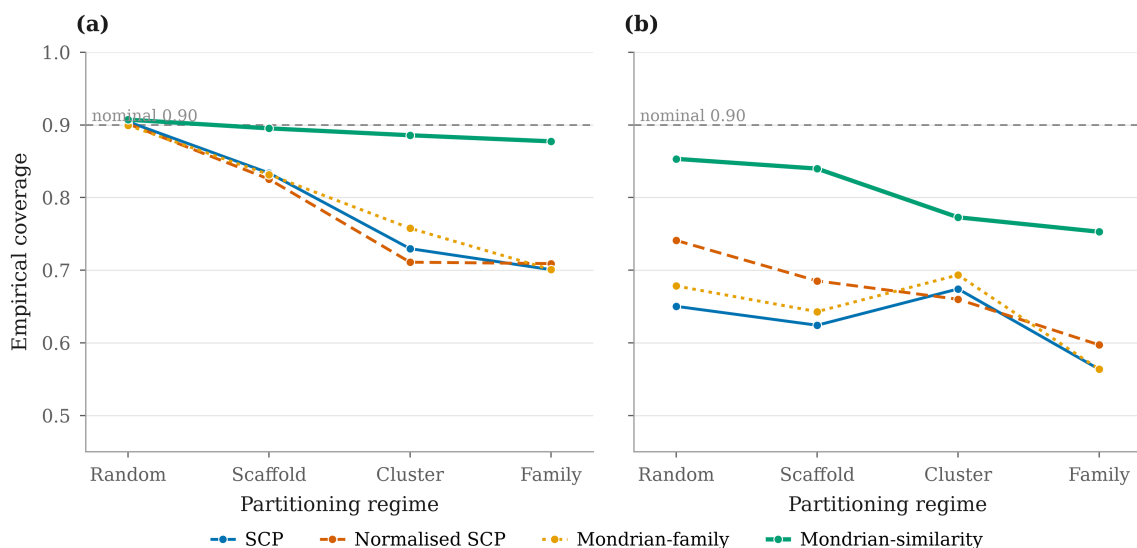


Figure 4. Coverage of nominally 90% conformal intervals. (a) Marginal coverage. (b) Coverage in the least-similar structural band. Line style encodes method alongside colour, so the exact coincidence of split conformal and the family-conditioned quantile under family holdout remains visible.

Marginal coverage also conceals a failure that is present even when it holds. On random splits, where overall coverage is a healthy 0.904, coverage in the least-similar structural band is 0.650 (Figure 4, right; Figure 5). A nominally 90% interval covers two-thirds of the repeat units that a screening campaign would actually be evaluating. A model can pass every validity check in current practice and still be wrong about a third of the unfamiliar polymers it is deployed on.

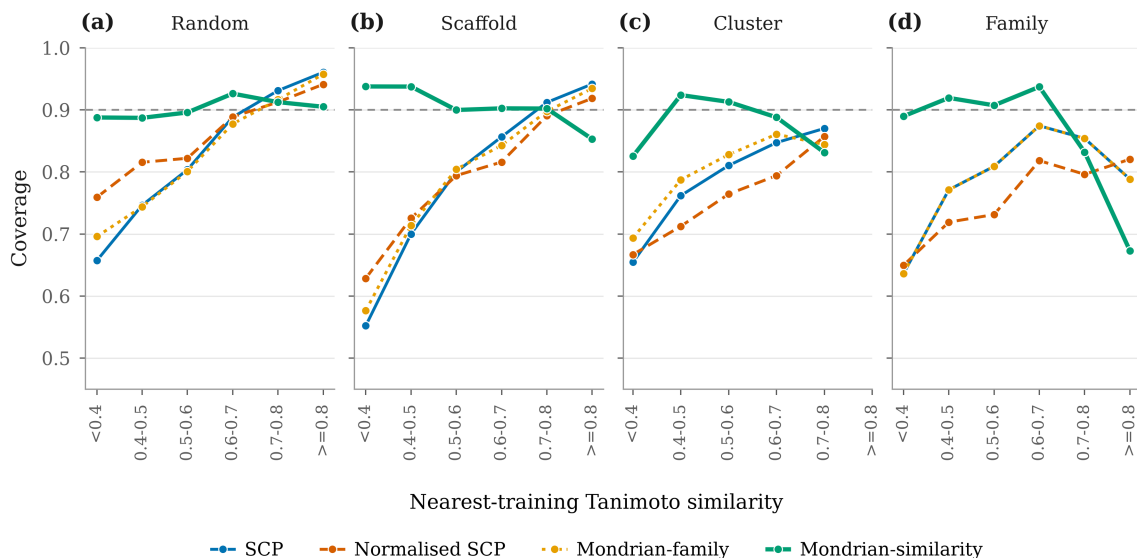


Figure 5. Conditional coverage against nearest-training Tanimoto similarity, one panel per split regime. Split conformal degrades monotonically with structural novelty; the novelty-conditioned quantile stays near nominal.

Neither obvious remedy works. The normalised predictor, which scales residuals by a fitted difficulty estimate, improves the worst band on random splits (0.741 vs 0.650) but fails with everything else under shift (0.825 scaffold, 0.711 cluster, 0.709 family) and is the *worst* of the four under cluster holdout. Conditioning the quantile on polymer family fails for a structural reason rather than an empirical one: under family holdout the withheld family has no calibration members by construction, so the predictor falls back to the global quantile for 100% of test structures and reproduces split conformal exactly (0.701, identical interval widths).

Conditioning on nearest-training similarity is the variant that survives, because that coordinate is defined for every repeat unit whether or not its family was seen. It holds 0.907, 0.895, 0.886 and 0.877 across the four regimes, never requiring a fallback under family holdout. On random

splits it achieves this while being *narrower* than split conformal (129.1 K vs 132.3 K), and it widens only where the chemistry demands it — to 155 K under scaffold, 232 K under cluster and 208 K under family holdout. That adaptivity is what the coverage-width criterion rewards (Figure 6): under family holdout, 411 against 48,020 for split conformal.

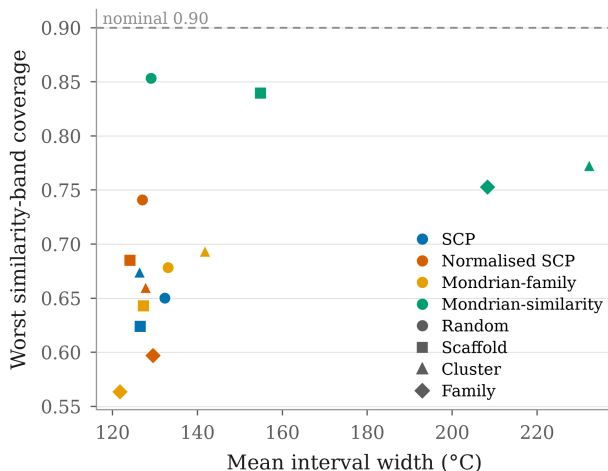


Figure 6. Worst similarity-band coverage against mean interval width. Marker colour is the conformal method, shape the split regime.

Whether width alone could have bought the same result is worth testing directly, because if it could the taxonomy would be doing no work. Under family holdout the width-matched control of Section 2.7 — split conformal inflated by the single constant $k = 1.67$ that equalises mean width with the novelty-conditioned predictor — reaches 0.908 pooled marginal coverage against the novelty-conditioned predictor’s 0.898. Marginally, therefore, a global inflation does match and slightly exceed it. Conditionally it does not: in the least similar band (Tanimoto < 0.4) the width-matched control covers 0.849 against 0.890. Recovering that band by global inflation alone requires $k = 1.93$, which carries the mean interval to 234 K against the novelty-conditioned 203 K, over-covering the bands that are already adequate in order to reach the one that is not. Two caveats attach to this control and both weaken the inflation strategy further. It is an oracle: k is computed from the test-set widths it is being compared against, so neither 1.67 nor 1.93 is knowable at prediction time. And the factor is regime-dependent, so a constant chosen on one holdout has no claim on another. The defensible statement is the conditional one: no single global inflation of split conformal reaches nominal coverage in the least-familiar band without over-covering everywhere else, and the factor that would be required cannot be computed when the prediction is made.

3.5. An applicability domain, and an honest limit on the repair

Resolving coverage by similarity band under family holdout (Table 5) turns the result into an operational rule. Split conformal covers 0.636 of structures whose nearest training analogue lies below Tanimoto 0.4, and 0.771, 0.809 and 0.874 in the next three bands. Its intervals are essentially constant in width (121–123 K) across all six bands, while the absolute error across those bands varies by a factor of 1.83, from 55.5 K in the least-similar band to 30.3 K in the best (Figure 7). Applying one width to a 1.8-fold spread in error is precisely the failure mode. Novelty-conditioned calibration instead holds 0.890, 0.919, 0.907 and 0.937 in the four least-familiar bands, at widths that scale from 246 K down to 149 K.

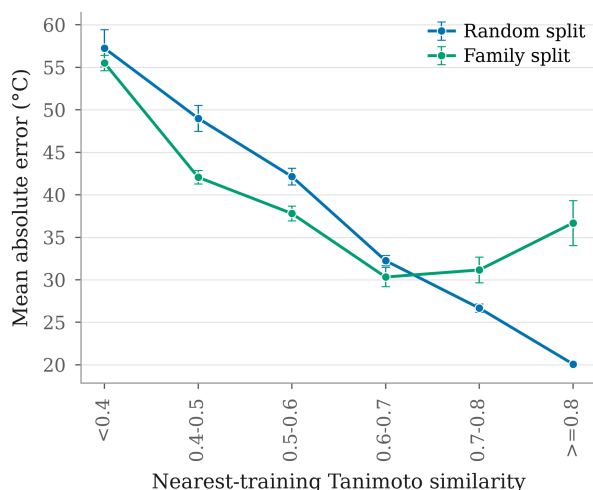


Figure 7. Mean absolute error against nearest-training similarity, for random and family-holdout splits. Group sizes are annotated.

The repair has a cost that should be stated plainly. In the most-similar band (Tanimoto ≥ 0.8) the novelty-conditioned predictor *under-covers*, at 0.673 against split conformal's 0.788, because it narrows those intervals to 88 K. Under a family holdout a test structure that closely resembles training data is unusual, and the calibration points populating that band come from families where high similarity genuinely did signal an easy prediction. The trade is nonetheless favourable in practice: that band holds 156 of 7,174 held-out predictions (2.2%), while the two least-similar bands hold 4,647 (64.8%). The recommendation is therefore to report novelty-conditioned intervals together with the nearest-training similarity itself, so that a reader can see which regime a given prediction sits in rather than trusting a single interval uniformly.

3.6. The penalty follows which structures were removed, not how many

The matched design separates the two effects that a conventional family holdout confounds (Figure 3, Table 6). Across twenty-one families and three repeats, with the test set held identical and the two comparison pools held identical in size:

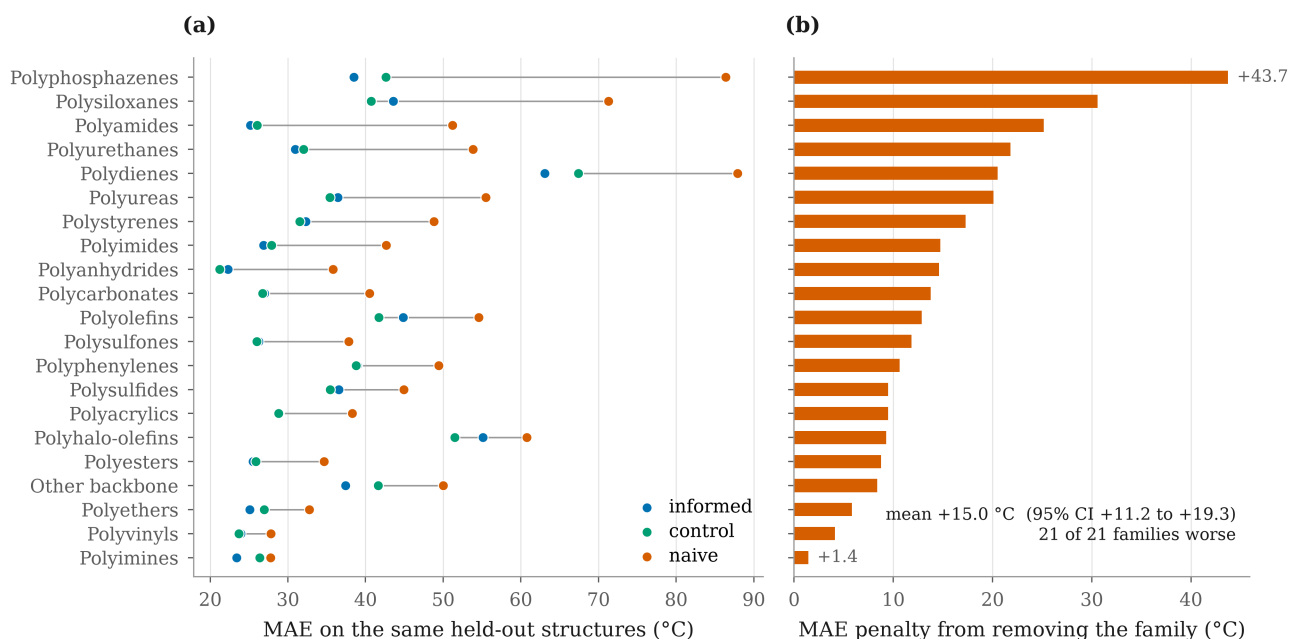


Figure 3. Matched-pair design. (a) The three training arms predicting an identical set of held-out structures; informed and control are almost coincident, while naive is displaced to the right in every family. (b) The family effect, naive minus control, with training-set size held constant.

- Removing an equal quantity of *unrelated* structures — the **control** arm —

changes MAE by +0.29 K (95% CI [-0.63, +1.24], Wilcoxon $p = 0.71$, worse in 9 of 21 families) for extremely randomised trees, and by -0.58 K (95% CI [-1.81, +0.64], $p = 0.45$, worse in 10 of 21) for histogram gradient boosting. The two models do not agree even on the sign.

- Removing *the family itself* — the **naive** arm — costs +14.97 K

(95% CI [+11.21, +19.28]) for extremely randomised trees and +15.15 K (95% CI [+11.02, +20.03]) for histogram gradient boosting, and **is worse in all twenty-one families for both models**. With forty-two of forty-two signs positive the Wilcoxon test returns its floor, $p = 9.5 \times 10^{-7}$ per model; the sign count is the result, not the p -value.

The two effects are therefore not merely different in size but different in character. The family effect is roughly 15 K and unanimous in sign across every family and both models. The size effect is under 1 K in magnitude, has opposite signs in the two models, and scatters around zero per family, from -3.6 K to +4.2 K for extremely randomised trees and from -8.9 K to +7.3 K for histogram gradient boosting. We deliberately do not quote a ratio between them: with the denominator's sign unresolved, a ratio is not a quantity. What the design establishes is that the total effect, +15.26 K and +14.57 K for the two models, is accounted for by the family term alone, and it does so on matched test structures, so it is not an artefact of families differing in intrinsic difficulty or in T_g spread.

What the two arms actually differ in is the *locality* of the removal, not its volume: the naive arm deletes a connected region of structure space, the control arm a diffuse sample of the same size scattered across the other twenty families. The chemical reading of that contrast is developed in Section 4.2; the measurement is the locality contrast.

Family-level magnitudes rank as the chemistry predicts. For extremely randomised trees, polyphosphazenes suffer most (+43.7 K), followed by polysiloxanes (+30.6 K) and polyamides (+25.2 K); at the other end polyimines (+1.4 K), polyvinyls (+4.1 K) and polyethers (+5.8 K) are nearly unaffected, because the linkages that set their backbone mobility survive elsewhere in the training set. Note that polyamides rank third here despite being the second-largest family in the dataset (999 structures) — further evidence that abundance does not confer transferability when the chemistry is distinctive.

The practical reading follows from the locality contrast. At a fixed training-set size, a pool that excludes the target family predicts it about 15 K worse than a pool of the same size that retains the family and excludes an equivalent quantity of unrelated structures instead. Either the family must be represented in training, or the prediction must carry an interval that widens honestly when it is not — which is the case for novelty-conditioned calibration made in Section 3.4.

3.7. The gap is not an artefact of the descriptor set

A natural objection is that the generalisation gap reflects a limitation of RDKit descriptors rather than of the data, and that a representation better suited to novel chemistry would close it. Repeating the random and family-holdout regimes with binary Morgan fingerprints (radius 2, 2,048 bits) in place of the descriptor block tests this directly, since the two representations share almost nothing: one is a set of physically motivated aggregate quantities, the other a sparse record of local substructural environments. Both use extremely randomised trees.

The gap survives the substitution and is not narrower (Table 7). Fingerprints are worse in absolute terms in both regimes — 35.8 K against 28.5 K on random splits and 62.7 K against 49.6 K under family holdout — and their family-holdout penalty is larger in absolute terms (+26.9 K against +21.1 K) while being indistinguishable as a ratio (1.752 against 1.738). Substructural environments offer no protection here in aggregate.

The per-family picture is not uniform, and one plausible mechanism is ruled out by it. Fingerprints are worse than descriptors on eighteen of the twenty-one withheld families, but they are *better* on three, and the largest single improvement is on polyphosphazenes (62.7 K against 87.1 K), the family whose absent P=N bits would be the obvious reason for a fingerprint

model to fail. Nor are they better on polysiloxanes (77.6 K against 70.4 K). An explanation in terms of missing backbone bits therefore does not survive contact with the per-family data, and we do not offer one. What the ablation shows is narrower and still useful: the gap is not specific to the descriptor block, so it cannot be dismissed as an artefact of that feature set. Whether some other representation — a model pretrained on millions of hypothetical polymers, for instance — could narrow it is untested here and is the natural next step (Section 4.5).

3.8. The calibration set is a second lever, with a caveat

Sections 3.4 and 3.5 repair conditional coverage by changing how the conformal *quantile* is computed. A second possibility is to leave the quantile alone and change what the calibration set contains. Standard split conformal draws calibration points at random from the training pool, so every calibration residual describes an interpolation — the model had relatives of that repeat unit in training. Under family holdout the test residuals describe extrapolation, so the quantile is calibrated against the wrong population. Rebuilding the calibration set by holding families out *within* the training pool produces residuals of the kind the test set will actually demand.

Both levers work, and to a similar degree (Table 9). Against a nominal 0.90 under family holdout, with the point model identical in all four arms:

Calibration set	Quantile	Coverage	Worst band	Width	<i>n</i> calibration
Random	Global (SCP)	0.701	0.564	122 K	1,366
Random	Novelty-conditioned	0.877	0.753	208 K	1,366
Family-out	Global (SCP)	0.855	0.753	187 K	3,349
Family-out	Novelty-conditioned	0.863	0.778	189 K	3,349

Rebuilding the calibration set lifts plain split conformal from 0.701 to 0.855 without touching the quantile rule, the model or the representation. It does so by widening intervals from 122 K to 187 K, which is the appropriate response: the residuals it now calibrates against are genuinely larger. The construction is not, however, the only thing that changes in that swap. The family-out construction also enlarges the calibration set from a mean of 1,366 structures to 3,349, because residuals are pooled over the held-out training families rather than drawn once. A larger calibration set tightens the quantile's own sampling error, so the 0.701 → 0.855 improvement is attributable to the two changes jointly and this experiment cannot separate them.

The two repairs are largely substitutes rather than complements. Applying both gives 0.863 marginal coverage, no better than the novelty-conditioned quantile alone at 0.877, because they address the same root cause from opposite ends — that calibration residuals understate extrapolation error. The combination has the best worst-band coverage of the four arms, 0.778 against 0.753 and 0.753, but that margin is not resolved: across the twenty-one families the combination beats the novelty-conditioned quantile alone on worst-band coverage in only 9, with a mean difference of +0.025 and a Wilcoxon *p* of 0.40. It should be read as no worse rather than as better.

The practical guidance follows the cost. Novelty-conditioned calibration is free: it reuses the same fitted model and calibration set, requiring only that similarity be computed. Family-out calibration requires refitting the model once per held-out training family — here eight additional fits per split. For most purposes the novelty-conditioned quantile alone is the better trade.

3.9. An additive baseline deteriorates less and predicts worse

The additive schemes that machine-learned models propose to replace make a different inductive assumption: that T_g is a composition-weighted sum of group contributions, with no interaction terms and no dependence on which linkages the groups sit in. If the family-holdout penalty is really about absent linkage chemistry, a model that never learns linkage-specific behaviour in the first place should lose less when a linkage type is withdrawn. It does, and it pays for that in accuracy everywhere (Table 11).

The group-contribution baseline of Section 2.6 gives 43.8 K MAE on random splits, 43.8 K under scaffold holdout, 54.9 K under cluster holdout and 59.8 K under family holdout. The absolute penalty, random to family, is +16.0 K against the ensemble's +20.0 K, and the ratio is 1.36 against 1.71. The flat random-to-scaffold step is the sharper observation: 43.84 K to 43.80 K, no change at all, where the ensemble loses 5.7 K. A model whose only inputs are group counts is indifferent to whether it has seen the scaffold before.

The honest way to read this is absolute first. The baseline is worse than the ensemble in every regime, by 15.7 K on random splits and by 11.7 K under family holdout, and its smaller penalty ratio partly reflects a worse starting point: it has less accuracy to lose. Resolved by family, it beats the ensemble on only two of the twenty-one withheld families, polysiloxanes (68.2 K against 69.7 K) and polysulfones (35.3 K against 36.7 K), both narrowly. It is not an alternative to the ensemble for prediction.

It is nonetheless independent support for the mechanism, from a model class with a different inductive bias and a different failure mode. Compositional additivity buys transferability at the cost of accuracy; the ensemble makes the opposite trade, and the size of the trade is what Sections 3.6 and 4.2 are about.

3.10. Sensitivity to the family taxonomy

Every family-dependent result inherits the labels, so the taxonomy is a legitimate target for a referee. During this work a boundary-split bug was found and fixed in the classifier: because the backbone was taken as the path between the two attachment points, a repeat unit written cut at its own characteristic linkage lost that linkage, so nylon-6 written `*NCCCCC(=O)*`` read as a plain hydrocarbon chain and was labelled a polyvinyl. The dimer construction of Section 2.2 is the fix, and the same revision replaced a catch-all hydrocarbon assignment with the Other backbone abstention class. Together these reassigned 204 of the 7,174 structures, 2.8% of the dataset, the largest movements being 88 polyesters that are really polycarbonates, 46 polyolefins carrying a backbone heteroatom, 23 polyethers that are polyesters, and 15 polyvinyls that are polyamides.

Rerunning the whole pipeline against both label sets gives a direct sensitivity measurement. The random, scaffold and cluster results are unchanged to three decimal places for every model, as they must be, since those regimes never consult a family label; pooled R^2 in those three regimes is likewise identical to three decimals. Under family holdout, where the labels do enter, pooled R^2 moves by at most 0.005 across the five models and is unchanged at 0.746 for histogram gradient boosting. The remaining family-dependent results move by about a kelvin and no conclusion changes sign: family-holdout MAE goes from 46.8 K to 48.1 K for histogram gradient boosting and from 48.9 K to 49.6 K for extremely randomised trees, and the matched-design family effect from +14.61 K to +14.97 K and from +13.28 K to +15.15 K. The one qualitative change is in the paper's favour. Under the old labels the family effect was positive in 20 of 20 families for one model but only 18 of 20 for the other, the two exceptions being polycarbonates and polysulfides; polycarbonates was also the family most altered by the correction, nearly doubling from 98 to 189 structures as the misassigned polyesters returned to it. Under the corrected labels the effect is positive in 21 of 21 families for both models. The earlier anomaly was a labelling error, not a counterexample.

This is a sensitivity check against one specific perturbation, not a proof of robustness to any taxonomy. A chemically different but defensible scheme — one that split polyimides by dianhydride class, say — would move the family-level numbers again. What the check establishes is the order of magnitude of that movement for a 2.8% relabelling: about a kelvin on aggregate errors, with the sign structure of the matched design intact.

4. Discussion

4.1. What a T_g leaderboard is actually measuring

The most consequential number in this study is not an error but a contrast. Changing the model changes MAE by 0.38 K between the two best and by 2.74 K across all four non-trivial regressors; changing the partitioning protocol from random to family holdout changes it by 20.0 K. Any comparison of architectures on this dataset that does not fix and report the split regime is dominated by a variable it does not control.

This reframes a decade of incremental gains. Reported improvements from kernel methods to tree ensembles to graph networks to pretrained chemical language models are real, but they are improvements in interpolation, measured where structural analogues are plentiful. None of the comparisons establishes that the newer model extrapolates better, because none was evaluated under a regime that requires extrapolation. That is not a criticism of the models; it is a criticism of the protocol they were scored under, and it is cheap to fix.

4.2. Why chemistry and not volume

The matched design measures a locality effect: at fixed training-set size, deleting a coherent chemical neighbourhood costs about 15 K on that neighbourhood, deleting a diffuse sample of the same size costs nothing that survives a change of model. The chemical reading of that asymmetry is the straightforward one. Descriptor-based models of T_g learn how particular backbone linkages translate into chain stiffness and interchain interaction, and those mappings are local to a linkage type. Remove every phosphazene, and no amount of additional polyester data teaches the model what a P=N backbone does to segmental mobility; it extrapolates organic intuition into inorganic chemistry and overpredicts those chains by 84 K on average (Section 3.3). Three lines of evidence support that reading: deterioration correlates with structural isolation while no size effect is resolved (Section 3.3); the withheld-family errors are systematic displacements of a consistent sign, about half the error on a typical family (Section 3.3); and an additive model that never learns linkage-specific behaviour deteriorates proportionally less when a linkage type is withdrawn (Section 3.9).

The consequence for data-collection strategy needs stating carefully, because the experiment removes data rather than adding it. What was tested is that withdrawing a median of 1.85% of the training pool at random, from chemistry already covered, does not measurably change accuracy on a held-out family, while withdrawing the same quantity as a coherent family does. That says a marginal fraction of the existing pool carries little information about a new backbone. It does not establish that *adding* data within covered chemistry would be useless, which is a different experiment at a different scale and one these results cannot settle. The defensible version of the recommendation is the comparative one: at equal cost, structures from an uncovered family are worth more than structures from a covered one, and a screening programme that plans its data collection around coverage of the structure space rather than around volume is acting on what was measured.

4.3. Why novelty-conditioning works and family-conditioning cannot

The contrast between the two Mondrian taxonomies is a structural fact rather than an empirical accident, and it generalises beyond this dataset. A conditional conformal predictor can only offer category-specific validity for categories it has calibration data for. Polymer family fails this test in exactly the deployment scenario of interest: the family is new, so it has no calibration members, and the predictor silently degrades to the unconditional one, here for 100% of test structures. Nearest-training similarity is computable for any repeat unit against any training set, its value is available before the label is, and its bands are populated by calibration structures drawn from every family, so the taxonomy remains well defined precisely where the family-based one collapses. The same argument recommends similarity — or any other deployment-computable novelty coordinate — over class-based taxonomies for conditional conformal prediction in materials problems generally, wherever the classes of interest are the ones the model has not seen.

The similarity taxonomy is not, however, free of the same defect in kind. A similarity band can also be empty of calibration members, and under cluster holdout it is: 12.8% of test structures fall back to the global quantile there, against 21.5% for the family taxonomy. The difference is one of degree and of prognosis. A family holdout empties the relevant family category by construction, every time, and no amount of additional calibration data fixes it; a similarity band is emptied only when the calibration pool happens to contain nothing that novel, which calibration data drawn more widely can in principle remedy. The advantage is real but it should be stated as a smaller and repairable blind spot rather than as none.

Section 3.4 also tested whether width alone explains the advantage. Marginally it can: an oracle inflation of split conformal matched to the same mean width reaches 0.908 against 0.898. Conditionally it cannot, missing the least-similar band at 0.849 against 0.890, and the global factor that would close that band costs 234 K of mean width against 203 K and is not knowable at prediction time. The claim we make is therefore about allocation, not about total width: the novelty-conditioned predictor puts its width where the error is, and no constant multiple of split conformal does that.

4.4. Practical recommendations

For anyone building or reporting a T_g model intended for screening:

- **Report the split regime as a primary result, not a methods detail**, and

report at least one chemistry-aware regime alongside any random split. The gap between them is the honest estimate of screening performance.

- **Repeat every regime that can be repeated, and say what the spread means.**

Scaffold and cluster splits vary by ± 8 –10 K across repeats. A family holdout is run once per family, so its ± 15 K is dispersion between families, not run-to-run noise; both are worth reporting, and conflating them is not.

- **Report errors resolved by polymer family, with the sign of the error.** An

aggregate MAE of 48 K conceals a range from 30 K to 85 K, and a user working on polysiloxanes is served by the wrong number; the -64 K bias on that family tells them more than either.

- ****Ship intervals, and condition them on a deployment-computable novelty**

coordinate.** Report the nearest-training similarity next to each prediction so a reader can see which regime it belongs to.

- **Fit every data-dependent preprocessing step inside the fold**, including

descriptor filtering. It is a one-line change to a pipeline and removes an unquantified optimism from the reported numbers.

4.5. Limitations

The target itself is noisy in a way no structural model can address. Independent literature reports of the same repeat unit differ by up to 115 K in this dataset, because a repeat-unit graph cannot express molecular weight, tacticity, crosslink density, thermal history or measurement protocol. A 28 K random-split MAE should be read against that floor, and part of the residual is irreducible.

The family taxonomy is our own. Section 3.10 measures its sensitivity to one 2.8% relabelling and finds about a kelvin of movement with every conclusion intact, but that is one perturbation, not a survey of chemically defensible alternatives. The validation in Table S1 should also be described accurately: it is a pinned regression suite, and its twenty-five cases are the ones the classifier's patterns were written to handle. It protects against future regressions and it demonstrates that the rules encode the chemistry we claim they encode; it is not an independent held-out validation set, and it cannot be read as an accuracy estimate for the taxonomy on unseen chemistry.

The matched design removes data rather than adding it, and removes a median of 1.85% of the pool in the control arm (Section 2.5). Its conclusions are about where a fixed data budget should be spent, not about the returns to enlarging it.

Our models are descriptor-based ensembles and one additive baseline. We show in Section 3.7 that the generalisation gap survives a change of representation to Morgan fingerprints, but we have not tested a pretrained chemical language model or a graph neural network. Such a model, having seen millions of hypothetical polymers during pretraining, might have effective coverage of families absent from this dataset's *labelled* portion. Testing whether pretraining narrows the family effect is the natural next experiment, and the matched design transfers to it unchanged.

Finally, the novelty-conditioned predictor under-covers the most-similar band under family holdout (Section 3.5) and falls back to the global quantile for 12.8% of structures under cluster holdout (Section 4.3). The affected groups are small and the direction of the trade favours screening, but the method is an improvement rather than a solution, and conditional coverage should continue to be reported rather than assumed.

5. Conclusions

Machine-learned glass transition temperatures are far more reliable inside familiar backbone chemistry than outside it, and standard practice measures only the former. On 7,174 experimental repeat units, moving from random to chemistry-aware partitions raises the error of the best model from 28.1 K to as much as 48.1 K and lowers pooled R^2 from 0.87 to 0.75. That 20.0 K shift is roughly fifty times the difference between the best and second-best model and seven times the spread across all four non-trivial regressors.

A matched-pair design shows that the penalty follows the locality of what is removed rather than its quantity: deleting an equal number of unrelated structures moves MAE by under 1 K, with the two models disagreeing on the sign, while deleting the family costs about +15 K and does so in all twenty-one families for both models. Transferability tracks how structurally isolated a family is; family size is not resolved at this sample size, and the two are nearly orthogonal, so the first result is not the second in disguise. An additive group-contribution baseline, which never learns linkage-specific behaviour, loses proportionally less across the same regimes and predicts worse everywhere, which is the same mechanism seen from a model class with a different inductive bias.

The uncertainty attached to these predictions fails in the same place. Split conformal intervals hold their nominal 90% marginally on random splits while covering only 65% of the least familiar structures, and lose marginal validity under shift. Conditioning the calibration on polymer family cannot repair this, because a withheld family has no calibration data by construction. Conditioning on nearest-training structural similarity does, holding 0.88–0.91 across every regime tested and allocating width to the bands where the error actually is. Rebuilding the calibration set from within-training family holdouts is a second lever that reaches 0.855 with the plain global quantile, though in this experiment it also enlarges the calibration set from 1,366 to 3,349 structures, so construction and size cannot be separated; combining the two levers is no better than the novelty-conditioned quantile alone.

The practical implication is a short list: report a chemistry-aware split, repeat what can be repeated, resolve errors by family and by sign, and attach novelty-conditioned intervals together with the similarity value itself. None of these is expensive, and together they turn a T_g model from something that scores well into something a screening campaign can act on.

Supplementary Materials

The following are available in the repository named in the Data Availability Statement: Table S1, the twenty-five reference polymers with expected and assigned family; Table S2, family-holdout deterioration against structural isolation, T_g shift, T_g interquartile ratio and family size, with

Pearson and Spearman correlations; Table S3, attachment-point counts per structure; Table S4, per-band coverage under each calibration construction; Table S5, absolute error resolved by nearest-training similarity band with the worst-to-best band ratio; Table S6, per-family group-contribution error under family holdout.

Author Contributions

Conceptualisation, [AUTHOR]; methodology, [AUTHOR]; software, [AUTHOR]; validation, [AUTHOR]; formal analysis, [AUTHOR]; investigation, [AUTHOR]; data curation, [AUTHOR]; writing—original draft preparation, [AUTHOR]; writing—review and editing, [AUTHOR]; visualisation, [AUTHOR]. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

All code, the curated structure-level table and every result file underlying this work are available at <https://github.com/Rugvedhs/spectrasense>, in the directory ``polymer-tg-benchmark``. The underlying experimental T_g records are redistributed with the POINT² benchmark [Xu2025] and derive from PoLyInfo [Otsuka].

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Conflicts of Interest

The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

T_g	glass transition temperature
MAE	mean absolute error
RMSE	root mean squared error
CI	confidence interval
SCP	split conformal prediction
CWC	coverage-width criterion
SMILES	simplified molecular-input line-entry system
PSMILES	polymer SMILES, with attachment points marked
SMARTS	SMILES arbitrary target specification
DSC	differential scanning calorimetry

T_g	glass transition temperature
DMA	dynamic mechanical analysis

Appendix A. Result tables

Point accuracy by split regime and model

regime	model	n splits	mae mean	mae sd	rmse mean	r2 mean
random	hist_gbr	10	28.131	1.015	41.037	0.865
random	extra_trees	10	28.513	0.931	41.656	0.861
random	random_forest	10	30.302	1.000	43.329	0.850
random	svr	10	30.875	0.896	45.482	0.835
random	median	10	93.786	1.650	112.146	-0.00338
scaffold	hist_gbr	10	33.807	7.764	45.950	0.792
scaffold	svr	10	35.336	6.152	49.328	0.761
scaffold	extra_trees	10	35.498	8.483	47.937	0.774
scaffold	random_forest	10	36.950	8.367	49.580	0.758
scaffold	median	10	95.709	13.666	114.607	-0.291
cluster	hist_gbr	10	44.466	10.373	58.590	0.538
cluster	extra_trees	10	48.705	11.271	62.946	0.462
cluster	random_forest	10	48.791	11.807	63.327	0.458
cluster	svr	10	53.943	17.063	69.875	0.329
cluster	median	10	121.607	24.225	138.638	-1.737
family	hist_gbr	21	48.107	15.393	62.676	0.256
family	extra_trees	21	49.563	16.186	63.836	0.226
family	random_forest	21	50.303	15.656	64.706	0.204
family	svr	21	53.062	18.278	67.953	0.094
family	median	21	95.456	28.605	111.516	-1.490

Conformal coverage, width and subgroup validity

regime	conformal	coverage	worst similarity coverage	mean width	fallback
random	Mondrian-family	0.899	0.678	133.078	0.076
random	Mondrian-similarity	0.907	0.853	129.075	0
random	Normalised SCP	0.901	0.741	126.994	0
random	SCP	0.904	0.650	132.327	0
scaffold	Mondrian-family	0.832	0.643	127.201	0.099
scaffold	Mondrian-similarity	0.895	0.840	154.832	0
scaffold	Normalised SCP	0.825	0.685	124.083	0
scaffold	SCP	0.834	0.624	126.459	0
cluster	Mondrian-family	0.758	0.693	141.777	0.215
cluster	Mondrian-similarity	0.886	0.773	232.226	0.128
cluster	Normalised SCP	0.711	0.660	127.744	0
cluster	SCP	0.730	0.674	126.358	0
family	Mondrian-family	0.701	0.564	121.721	1.000
family	Mondrian-similarity	0.877	0.753	208.335	0
family	Normalised SCP	0.709	0.597	129.539	0
family	SCP	0.701	0.564	121.721	0

Matched-pair effects, with bootstrap confidence intervals

model	effect	n families	mean	ci low	ci high	wilcoxon p	n families positive
extra_trees	family_effect	21	14.971	11.207	19.284	9.54e-07	21
extra_trees	size_effect	21	0.292	-0.630	1.236	0.708	9
extra_trees	total_effect	21	15.263	11.351	19.829	9.54e-07	21
hist_gbr	family_effect	21	15.145	11.018	20.033	9.54e-07	21
hist_gbr	size_effect	21	-0.580	-1.807	0.644	0.452	10
hist_gbr	total_effect	21	14.565	10.628	19.346	9.54e-07	21

Family-holdout error and deterioration (histogram gradient boosting)

group	n	mae	r2	deterioration	mean nn similarity
Polyphosphazenes	118	84.839	-1.339	3.016	0.304
Polydienes	77	79.648	0.020	2.831	0.367
Polysiloxanes	209	69.701	-0.368	2.478	0.388
Polyureas	44	61.770	0.312	2.196	0.506
Polyolefins	126	61.626	0.418	2.191	0.397
Other backbone	58	52.457	-0.783	1.865	0.403
Polyhalo-olefins	56	51.667	0.141	1.837	0.333
Polyurethanes	189	47.890	-0.192	1.702	0.477
Polyamides	999	46.924	0.556	1.668	0.513
Polyphenylenes	423	46.370	0.712	1.648	0.463
Polyanhydrides	53	43.798	0.302	1.557	0.489
Polystyrenes	250	42.206	0.312	1.500	0.399
Polysulfides	60	42.020	0.746	1.494	0.516
Polyimides	1707	37.759	0.584	1.342	0.406
Polysulfones	232	36.654	0.565	1.303	0.561
Polyimines	52	36.418	0.225	1.295	0.532
Polyacrylics	661	36.249	0.382	1.289	0.412
Polyesters	761	34.929	0.728	1.242	0.521
Polycarbonates	189	33.745	0.741	1.200	0.558
Polyethers	713	33.362	0.816	1.186	0.553
Polyvinyls	197	30.217	0.497	1.074	0.430

Pooled R2 against the mean of per-split R2, by regime and model

regime	model	n predictions	r2 pooled	r2 mean of splits
random	median	14350	-0.00299	-0.00338
random	svr	14350	0.835	0.835
random	random_forest	14350	0.850	0.850
random	hist_gbr	14350	0.866	0.865
random	extra_trees	14350	0.862	0.861
scaffold	median	14863	-0.160	-0.291
scaffold	svr	14863	0.781	0.761
scaffold	random_forest	14863	0.777	0.758
scaffold	hist_gbr	14863	0.808	0.792
scaffold	extra_trees	14863	0.791	0.774
cluster	median	24985	-0.794	-1.737
cluster	svr	24985	0.514	0.329
cluster	random_forest	24985	0.623	0.458
cluster	hist_gbr	24985	0.679	0.538
cluster	extra_trees	24985	0.627	0.462
family	median	7174	-0.227	-1.490
family	svr	7174	0.618	0.094
family	random_forest	7174	0.715	0.204
family	hist_gbr	7174	0.746	0.256
family	extra_trees	7174	0.720	0.226

Group-contribution baseline against the ensemble, by regime

model	random	scaffold	cluster	family	penalty K	penalty ratio
group_contribution	43.845	43.803	54.860	59.800	15.955	1.364
hist_gbr	28.131	33.807	44.466	48.107	19.977	1.710

model	random	scaffold	cluster	family	penalty K	penalty ratio
median	93.786	95.709	121.607	95.456	1.671	1.018

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